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A systems thinking approach to operationalizing integrated water resources management in Ethiopia through leverage points and feedback loops

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Abstract

Operationalizing Integrated Water Resources Management (IWRM) in climate-vulnerable and water-insecure contexts is constrained by complex interactions among hydrological variability, infrastructural deficits, institutional capacity, and socioeconomic conditions. This study applies an integrated semi-quantitative systems framework to identify leverage points and dominant feedback loops that shape IWRM implementation in Ethiopia. This approach combines causal loop diagramming, eigenvector centrality, MICMAC influence-dependency analysis, and structural loop gain assessment to examine how the system structure governs behavior. Six leverage points, household income level, access to reliable water supply, governance effectiveness, economic water scarcity, food availability and access, and irrigation coverage, emerge as critical system connectors and strategic intervention nodes relevant to SDG 6.5 implementation. The results indicate that IWRM performance is driven by isolated sectoral action rather than by reinforcing feedback linking water access, livelihoods, food availability, and economic water scarcity, thus amplifying vulnerability to rainfall variability, limited storage capacity, and chronic underinvestment. The balancing feedback between governance effectiveness and infrastructure investment suggests a potential regulatory pathway to stabilize system performance, though its influence is constrained by institutional fragmentation and financial constraints. By explicitly linking system behavior to DG 6.5.1 (degree of IWRM implementation), this study demonstrates that effective IWRM operationalization requires coordinated interventions that simultaneously strengthen water infrastructure, governance capacity, and livelihood resilience. The proposed framework complements SDG monitoring by revealing the structural constraints and feedback mechanisms that must be addressed to achieve integrated, equitable, and climate-resilient water management in Ethiopia and other similar water-insecure countries.

Keywords IWRM, System thinking, Leverage points, Feedback loops, Economic water scarcity, Water infrastructure, Water governance, Ethiopia



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1 Introduction

1.1 Barriers to integrated water resources management (IWRM) operationalization

Integrated Water Resources Management (IWRM) is widely recognized as a comprehensive and participatory approach for coordinating the development and management of water, land, and related resources to maximize economic efficiency, social equity, and environmental sustainability [1, 2]. In contrast to the traditional fragmented, sector-specific water management approach, IWRM explicitly emphasizes cross-sectoral coordination, stakeholder participation, and basin-scale planning across multiple spatial and temporal scales [1, 3, 4]. Sustainable Development Goals 6, Target 6.5, stipulates that by 2030, integrated water resources management will be implemented at all levels, including through transboundary cooperation, as appropriate.

However, the implementation of integrated water resources management (IWRM) in developing countries is constrained by limited infrastructure, weak institutional capacity, fragmented governance, and insufficient financing and management systems, which require pragmatic, context-adapted approaches [5, 6]. Consequently, there is increasing recognition of adaptive management approaches that emphasize flexibility and context-sensitive strategies [5, 7, 8]. Ethiopia has exemplified these challenges. Although IWRM principles are incorporated into national water management policies, implementation is constrained by complex hydrology, low water storage capacity, underdeveloped hydraulic infrastructure and distribution networks, limited budget allocations, and fragmented institutional responsibilities, which increase environmental and social trade-offs [9–13]. Weak stakeholder awareness, poor coordination, and insufficient regulatory enforcement have also resulted in unmet demands, inequitable access, inefficient water use, and underutilization of water and land resources [12, 14–16].

The climate-induced variability in rainfall further complicates these barriers. Recurrent droughts in lowland regions and floods in the highlands [10, 17, 18] destabilize water availability, damage infrastructure, accelerate sedimentation, and undermine rain-fed agriculture [19, 20]. Limited water storage capacity, high evaporation losses, inefficient distribution systems, and underdeveloped hydraulic and irrigation infrastructure change hydrologic variability into economic water scarcity, restricting access to reliable water supplies and reinforcing food insecurity and poverty [10, 21, 22]. Simultaneously, low investment in modern technology, weak data infrastructure, integration with a weak monitoring and management system, limited adoption of modern technologies, and inadequate maintenance have reduced efficiency in water allocation and use and constrained planning, coordination, and adaptive management [10, 23, 24]. Consequently, water resources are often mismanaged, with underutilized reservoirs, inefficient irrigation systems and coverage, and poor data collection, further aggravating economic water scarcity, agricultural vulnerability, and food insecurity [10, 23, 24].

These challenges have been intensified by rapid population growth, urbanization, and agricultural and industrial expansion, resulting in deforestation, land degradation, pollution, and unsustainable land use. Inadequate waste management infrastructure and pollution control measures exacerbate these challenges by degrading water quality and quantity, and increasing competition among users [9, 25, 26]. Likewise, deforestation, land degradation, and unsustainable land use reduce water availability and increase sedimentation in reservoirs [27–29]. Poverty and weak market access also reinforce these

dynamics by limiting households' capacity to invest in irrigation, climate adaptation, and sustainable management [30–32].

These constraints differentiate Ethiopia's IWRM implementation challenges from traditional water management failures, as they arise from systemic interactions among hydrology, infrastructure, governance, and livelihoods rather than isolated technical limitations. Thus, the integrated nature of IWRM barriers requires an analytical approach capable of capturing feedback, interdependence, and non-linear relationships among system components [33, 34].

Thus, addressing such complexity requires an analytical approach that goes beyond fragmentary studies and explicitly captures feedback, leverage points, and dynamic relationships across sectors [33, 35, 36]. Systems thinking tools such as causal loop diagrams (CLD) are well-suited to this purpose as they facilitate the representation of dynamic interactions across hydrological, infrastructural, institutional, financial, and governance domains, rather than addressing isolated water management problems [33, 36–38]. Accordingly, this study applies an integrated system-thinking framework combining structural network analysis, MICMAC influence-dependence classification, and feedback loop analysis to identify leverage points and dominant feedback structures by evaluating structural relationships within Ethiopia's IWRM system [37, 39, 40] that constrain or enable IWRM operationalization in Ethiopia. This study aligns the IWRM assessment with SDG 6.5 while grounding recommendations in Ethiopia's policy and hydrological context, responding directly to calls for context-specific implementation-oriented IWRM analysis [5, 41].

1.2 Leverage point and dominant feedback loops as an instrument for IWRM operationalization

Identifying leverage points and dominant feedback loops is central to advancing system dynamics and is an important strategy for operationalizing IWRM. The leverage points were first introduced by Meadows [42] to describe the locations within a system where small, well-targeted interventions can produce significant changes in overall system behavior. Based on this, Forrester [43] and Sterman [58] developed system dynamics modeling as a method for representing complex feedback processes, enabling the analysis of non-linear interactions in social-ecological systems.

In structural system analysis, Godet [34] developed the MICMAC method to assess the influence-dependence relationships among systemic variables, which was expanded by Nabong [44] through the integration of social network analysis to identify key variables driving sustainable transitions. In contrast, Olive [37] applied graph theory to examine dominance and structural loops [7, 19]. Kampmann and Olive [40] formalized an analytical method for structural dominance analysis, providing a mathematical basis for identifying the feedback structure that applies the strongest control over system behavior. Oliva [45] further refined these techniques, emphasizing their utility in detecting dominant feedback loops that determine long-term system trajectories.

Recently, Schoenberg et al. [46] introduced the loops that matter (LTM) method, which quantifies the relative influence of feedback loops to explain model behavior. Hayward and Roach [47] compared various loop dominance measures to evaluate their interpretive consistency and applicability to policy analysis [46]. This method was

consolidated through the structural analysis of system dynamics models, integrating the dominance method to better link the system with behavior.

In the field of water resource management [33] system dynamics modeling was applied to identify leverage points and cascading impacts in the water–energy–food nexus systems. Endreny [48] used a system-based framework to locate leverage points in river basin restoration, whereas Huang et al. employed a system dynamics approach to model groundwater dynamics, emphasizing the role of feedback mechanisms in sustainable resource use [49]. Used participatory modeling workshops to identify feedback loops linking hydrological, human, and climatic systems. Theoretical contributions by [40, 50] emphasized loop-gain analysis as a quantitative measure of feedback strength, which is essential for determining which loops dominate the system behavior.

Building on these foundations, this study introduces an integrated framework that combines kumu.io, Vensim PLE, and Excel-based loop-gain analysis to identify leverage points and dominant feedback loops for IWRM Operationalization.

2 Methodology

This study employed systems thinking and semi-quantitative structural analysis to examine the interactions among barriers constraining the operationalization of IWRM in Ethiopia. Integrated network analytical techniques and structural feedback loop gain analysis were used to identify the relative leverage points and dominant feedback structure governing IWRM behavior, rather than to estimate statistical inferential effects. The analytical process integrates (a) the identified key variables (barriers) through literature review and validated by consulting the expert, (b) qualitative Causal Loop Diagram (CLD) (WWW.kumu.io and Vensim PLE), (c) quantitative MICMAC (Impact Matrix Cross-Reference Multiplication Applied to a Classification) and eigenvector centrality analysis using Kumu.io, and (c) structural loop-gain analysis (using Excel), adapted from [37, 40].

2.1 Identification of variables and construction of the CLD

System variables were identified and analyzed through a structured review of targeted peer-reviewed literature, in which IWRM challenges are framed as ‘barriers’; through a narrative review; and by integrating existing knowledge from peer-reviewed articles, government reports, policies, and background papers. The narrative review approach does not focus on strict inclusion criteria and quantitative data, allows for flexibility [51, 52] in addressing diverse evidence, and offers a comprehensive understanding of the country’s IWRM Operationalization barriers and their interactions. However, this research focused on studies published between 2006 and 2025, excluding background papers. Relevant papers were searched across academic databases, including Google Scholar, PubMed, ScienceDirect, Web of Science, Scopus, and government and institutional reports. The inclusion criteria targeted papers that discussed water management and development and integrated water resources management, using keywords and Boolean operators (AND, OR). Integrated water resources management in developing countries, adaptive water management, water resources management, integrated water resources management in Ethiopia, water governance, water infrastructure, and sustainable water management in Ethiopia. Variables were selected based on their impact on national-level IWRM performance, their appearance consistently across multiple

sources, and their direct influence on water availability, access, infrastructure, and technology performance, institutional capacity, socioeconomic outcomes, and investment. According to [43, 53], identifying variables from the literature ensures conceptual validity and alignment with established frameworks in system dynamics modeling. Overlapping variables were consolidated to ensure conceptual clarity and analytical traceability. All variables were defined using neutral, direction free terminology (governance effectiveness, investment in water infrastructure, access to reliable water supply etc.) in accordance with best practices in causal loop modeling.

Causal relationships and link polarities were first established based on evidence from the literature reviews and validate with in-person expert discussion (Table 1 of supplementary material), with qualitative assessment used to determine the relative strength of relationships.

Due to limited research funding, in-person workshops or focus group discussions could not be organized. Instead, a purposeful expert consultation approach was adopted for this study. Experts were selected from the Ministry of Water and Energy, regional water bureaus, zonal water offices, universities, research centers, and private consulting and construction firms in Ethiopia. The selection criteria included professional experience in water resources management and development, involvement in policy or implementation, study and design of water projects, research, and familiarity with Ethiopia's water resource management and development conditions. Most experts had more than ten years of relevant experience.

Six professional experts were consulted individually through in-person discussion to validate the key barriers, causal relationships, and link polarities initially identified and developed through an extensive literature review. A discussion guide with key themes and open-ended questions, derived from the literature, was used to ensure consistency across consultations. The discussions focused on the relevance of variables, direction of causal relationships, and link polarity rather than numerical calibration.

Prior to each discussion, the purpose of the study was clearly explained to the individual expert, and their voluntary agreement to participate was obtained before the discussion continued. All methods were carried out in accordance with relevant research ethics guidelines and regulations. The discussions focused solely on professional opinions regarding the identified barriers (variables) and causal links related to the implementation of integrated water resources management and development, and did not involve the collection of personal, sensitive, or biological data.

2.2 Causal loop diagram development and system boundary

The CLD was constructed by translating validated variables into directional causal relations based on documented cause-and-effect evidence from literature and expert consultation. Each causal link was assigned a polarity (positive or negative) to indicate the direction of influence. Model system behavior focuses on the underperformance of IWRM implementation in Ethiopia, particularly the emergence of water insecurity, livelihood vulnerability, and governance constraints. System boundaries were defined to include hydrologic variability, water infrastructure, governance and institutional arrangements, and livelihood-related socioeconomic dynamics at national and basin scales.

A CLD was constructed using www.kumu.io, an online platform for mapping and visualizing complex system interrelationships, to represent causal relations among variables, and Vensim PLE. Each variable was defined by its causal direction (direct or inverse) and relative influence on the other system components. These relationships were then modeled in Kumu.io to generate an interconnected visual system map. The resulting CLD (Fig. 2) enabled the identification of reinforcing (R) and balancing (B) feedback loops, based on theoretical and empirical evidence from the literature [53].

The expert reviewed the draft CLD and feedback loops to validate the causal relationship, feedback loops, and model structure. The finalized CLD was analyzed using eigenvector centrality and the MICMAC influence-dependence method to identify key leverage points and dominant loops within the system.

The absence of in-person workshops and focus group discussions may limit opportunities for the construction of an interactive model. However, iterative expert validation and triangulation with the literature were used to enhance the model's robustness and credibility.

2.3 Identifying leverage points using eigenvector centrality and MICMAC analysis

The final CLD was converted into an adjacent matrix to enable semi-quantitative structural analysis. To construct the CLD, a structural-metric analysis was performed to evaluate the relative importance of internal dynamics and the structural embeddedness of each barrier within the system. Eigenvector and MICMAC influence-dependence analyses were performed using Kumu.io software to evaluate the systemic importance of each barrier.

2.3.1 Eigenvector centrality of variables

Eigenvector centrality is a structural analysis method used to assess the relative structural embeddedness of a variable within a network, indicating its connectivity to other influential variables rather than its empirical effect size [37, 40]. Unlike degree centrality, which accounts for direct links, eigenvector centrality captures both direct and indirect influences through connections to other highly central nodes [33, 54]. Variables with high eigenvector centrality act as systemic leverage points because a change in their values ripples widely throughout the model [55].

In the system dynamics application of the IWRM, this metric helps identify relatively influential nodes (variables) that integrate environmental, hydrological, infrastructural, technical, socioeconomic, and governance subsystems [33, 35, 44]. For example, groundwater extraction rates or institutional governance parameters may exhibit high centrality, suggesting that policy adjustments could yield a system-wide effect [35, 56]. When combined with structural dominance analysis, eigenvector centrality enhances interpretability by linking loop strength to nodal importance, thereby providing a dual perspective on system leverage.

Thus, the key structural indicator used here is eigenvector centrality, defined as $Ax = \lambda x$, here $x = (E_1, E_1, \dots, E_n)^T$, where E_n is the eigenvector centrality of node i and λ is the largest eigenvalue of the adjacency matrix A , a_{ij} is Elements of the adjacency matrix of the system, A and it represents the connection from node i to j , E_j is Eigenvector centrality of node j , which is connected to node i , and Each component E_i represents the centrality of variable i , reflecting the extent to which it is influenced by other highly

connected variables [55, 57]. In system-dynamics terms, this measure captures a variable's potential to contribute to strong feedback behavior. Thus, eigenvector centrality was computed on the same network to quantify each node's relative prominence within the feedback structure [39, 55]. Eigenvector centrality assigns higher scores to nodes connected to other highly connected nodes, thereby enabling the detection of structurally central barriers.

Mathematically, for each variable i :

$$E_i = \frac{1}{\lambda} \sum_{j=1} a_{ij} E_j$$

2.3.2 MICMAC influence–dependence framework

MICMAC influence-dependence analysis decomposes structural relationships into direct and indirect influences to determine each variable's degree of driving power and dependence [34]. The MICMAC influence-dependency score was applied to classify the variables based on their relative influence and dependence, identifying potential leverage points within the model system configuration. According to [33, 34, 44], barriers were categorized into four groups. Figure 1 (1) Variables exhibiting high influence and low dependence are identified as drivers (protectors), representing potential leverage points and core structural drivers whose transformation could yield significant systemic changes. (2) Dependent (monitor) barriers (low influence, high dependence), representing outcome indicators of systemic performance, reflecting areas sensitive to change yet less capable of initiating transformation. (3) Linkage (target) barriers (high influence and low dependence), reflect areas sensitive to change yet less capable of initiating transformation, and (4) autonomous (incidental) variables (low influence and dependence) represent contextual or peripheral constraints that inhibit improvement without directly steering the system dynamics. This combined analysis provides a structural understanding of the systemic interactions that shape the IWRM performance in Ethiopia.

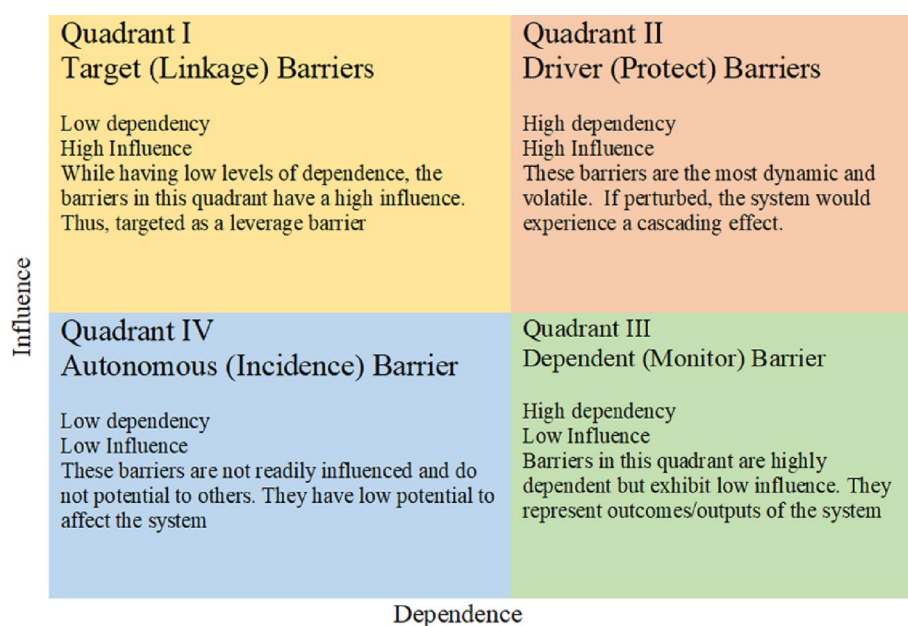


Fig. 1 Dependence chart to identify the role of barriers to IWRM operation (Adopted from [33, 34, 44, 59].

The MICMAC method [34, 58], which quantifies each variable's influence (M_i) and dependence (D_i) within a cross-impact matrix. M_i measures the structural leverage or driving power of variable i . Variables with high M_i and low D_i are identified as key drivers, and they are more likely to participate in dominant feedback loops.

Therefore, variables with the highest eigenvector centrality (E_i) values, along with MICMAC driver classifications (high influence and dependence), were identified as leverage points for the subsequent loop analysis.

2.4 Identifying dominant feedback loops through eigenvalue and MICMAC integration

Combining eigenvector centrality and MICMAC analysis creates a robust multi-criteria framework for identifying leverage points in an IWRM system. Eigenvector centrality locates structurally significant variables, whereas MICMAC positions them within an influence-dependence hierarchy [34, 44]. Integration helps distinguish between structurally dominant variables based on network position and strategically dominant variables based on systemic influence [59, 60]. emphasized that integrating analytical and participatory methods bridges the gap between system modeling and practical governance. By aligning model-based findings with stakeholder priorities, decision-makers can focus on interventions that not only have structural significance but are also socially and institutionally feasible. This dual framework enhances adaptive IWRM by identifying key policy interventions and revealing cross-sectoral interdependency, thereby increasing the resilience to uncertainty and change [33, 49].

2.5 Structural and system dynamics integration framework

After identifying the leverage points, their positions and roles within the CLD's feedback loops were analyzed. Each leverage point was traced across the reinforcing and balancing loops to understand its contribution to the persistence or mitigation of systemic barriers. Thus, feedback loops were automatically extracted from the CLD using the Loop Identifier function in *Vensim PLE*. Each loop is labeled as either reinforcing (R) or balancing (B) based on the product of the link polarities. For the structural dominance analysis, only feedback loops associated with at least one identified leverage point were retained.

Identifying dominant feedback loops by integrating eigenvector centrality and MICMAC analysis is theoretically grounded in both network centrality analysis and system dynamics [34, 37, 40]. Eigenvector centrality provides a mathematically rigorous measure of a variable's systemic importance by assessing its causal links with to other variables [33, 44]. A variable with a high eigenvalue score not only influences others but also participates in strongly interconnected feedback structures that amplify system behavior [33, 44].

The MICMAC framework [34] complements this by highlighting variables with strong structural leverage (high influence and low dependence), that act as system drivers. Combining both through $Wi = Ei + Mi$ captures the dual nature of dominance, dynamic centrality, and structural power. Summing across loop members approximates the collective structural gain of a loop [37, 50], and dividing by the loop length accounts for the principle that shorter feedback loops typically dominate because of reduced delays [40]. Normalization by the maximum value ensures comparability and allows the

calculation of a relative dominance index consistent with network and system-dynamics conventions.

Thus, to quantify the structural strength of each feedback loop, this study adopted the loop-gain formula originally proposed by [37] and extended by [40]. Each variable's weighted importance (W_i) combines its eigenvector centrality (E_i) and MICMAC influence (M_i). This multiplicative relationship combines dynamic centrality and structural leverage, ensuring that highly connected and influential variables carry a greater weight in identifying loop dominance.

$$W_i = E_i \times M_i$$

For a given feedback loop L_k with n_k member variables, the total structural gain LG_K is computed as

$$LG_K = \sum (E_i \times M_i) \quad \text{for all } i \in L_k.$$

Because longer loops inherently accumulate more contributions, a normalized mean value is used:

$$\overline{LG_K} = \frac{1}{n_k} \sum (E_i \times M_i)$$

This adjustment accounts for loop length and prevents bias towards larger loops [37, 40].

Loop scores were normalized to the [0, 1] range to enable comparisons

$$NLG_K = \left[\frac{\overline{LG_K}}{\text{Max}(\overline{LG_J})} \right].$$

Normalized loop gain (NLG_K) was used as the dominance index. Loop influence scores were normalized within each loop category by dividing the composite score of each loop by the maximum score observed in the category. Consequently, the highest-scoring loop in each category receives a normalized value of 1, even though the absolute average weights may differ across categories.

The feedback loops are ranked in descending order of NLG_K ; the loop with the highest value is considered the most dominant. Finally, this study introduces an integrated framework that combines Kumu.io, Vensim PLE, and Excel-based loop-gain analysis to identify leverage points and dominant feedback loops for IWRM Operationalization, focusing solely on either structural metrics or feedback dynamics. This framework links eigenvector centrality and MICMAC influence-dependence analysis from Kumu.io with qualitative feedback loop identification in the Vensim PLE. The resulting integration enables a semi-quantitative assessment of loop gain in Excel, bridging the structural influence and dynamic feedback strength. This approach provides a novel means of prioritizing interventions in water systems by underscoring the most influential variables and feedback mechanisms, without relying on a full-scale simulation model.

2.6 Limitations and interpretation scope

Despite its integrative strengths, the proposed framework has several limitations. Kumu.io effectively visualizes the system's structure and calculates influence metrics such as eigenvector centrality and MICMAC, but it cannot automatically detect dominant loops

or isolate all loop structures associated with specific variables. While Vensim is suitable for qualitative loop identification, it lacks the advanced simulation and decision-support functions found in professional versions. Furthermore, the Excel-based loop gain calculation is static and does not capture temporal dynamics. Consequently, while the approach identifies key leverage points and dominant loops, it does not simulate policy interventions or dynamic responses, underscoring the need for further validation with advanced system dynamics tools, such as WEAP or Vensim DSS.

The resulting metrics represented the relative behavior of the modeled system and were sensitive to variable selection, assumed causal direction, and exclusion of non-linearity and quantitative delays, which were interpreted as indicative of structural priority and intervention pathways rather than predictive probabilistic outcomes. In general, this integrated methodological approach provides a coherent framework for identifying leverage points and feedback mechanisms relevant to improving IWRM operationalization in Ethiopia.

3 Result

The results were derived from the structural and relational analyses. The analysis was based on eigenvector centrality, MICMAC influence-dependency positioning, and normalized loop-gain indices derived from the system's causal loop structure. Thus, the results indicate structural importance, relative influence, and dominance within the model system rather than statistically significant effects (hypothesis testing, confidence intervals, or p-values) in a probabilistic sense. Consequently, the reported ranking reflects comparative system behavior rather than the empirically estimated effect sizes or causal magnitudes.

3.1 Barriers to the IWRM model

Read arrows indicate inverse causal relationships, and the blue arrows indicate direct causal relationships.

Figure 2 illustrates a causal loop diagram (CLD) of operational barriers to IWRM implementation in Ethiopia, constructed in Kumu.io. The diagram captured the network of causal interactions among the identified barriers. In total, 142 direct causal connections were identified within the system. Among these, 117 connections exhibit direct relationships, whereby a change in one barrier produces a change in the same direction in connected barriers. The remaining 25 exhibited inverse relationships, in which a change in one barrier produced an opposite change in the connected barriers. The term "reinforcing and balancing" is used exclusively to characterize closed feedback loops and not individual causal links, indicating a causal link in which a change in one barrier leads to a similar change in the connected barrier, whereas 25 connections exhibit opposite directions.

3.2 Key barriers and roles in IWRM operations

Eigenvector centrality values indicate the positional importance of variables within the causal network, capturing the extent to which each variable is connected to other structurally influential variables. Barriers with high eigenvector centrality are network leaders, although they may not have the strongest local influence (www.kumu.io). The numerical scale of eigenvector centrality reflects normalization and relativity, and in

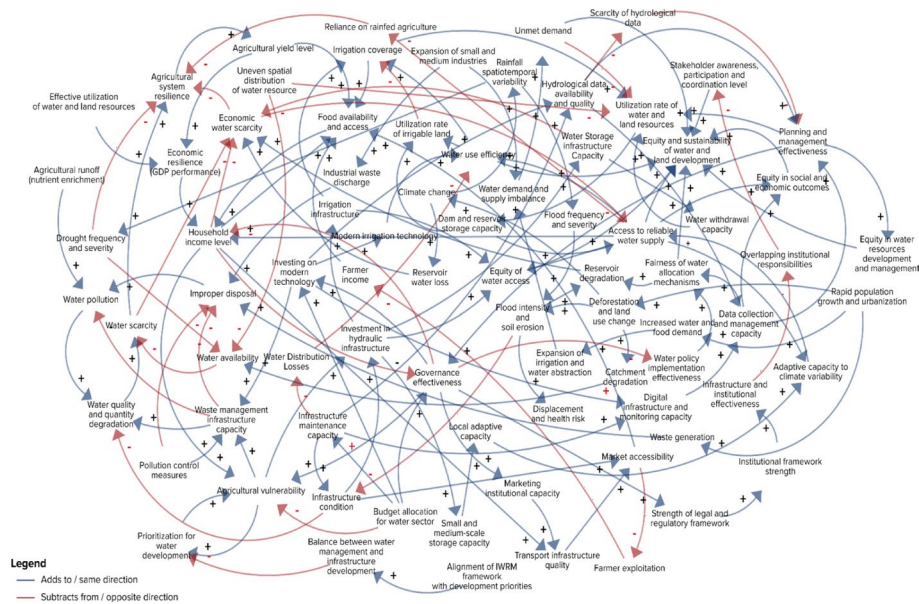


Fig. 2 The causal loop diagram developed for the barriers to IWRM operation in Ethiopia

Table 1 Results for the semi-qualitative analysis of the leader barriers to IWRM operation systems

Highly structurally embedded system leader	Eigenvectors centrality	Other key barriers involved in structurally dominant feedback loops
Household income level	0.062	Irrigation coverage
Investment in water infrastructure	0.058	Investment in hydraulic infrastructure
Equity and sustainability of water and land development	0.058	Water storage infrastructure capacity
Food availability and access	0.057	Water use efficiency
Economic water scarcity	0.044	
Transport infrastructure quality	0.037	
Waste management infrastructure capacity	0.035	
Equity of water access	0.035	
Agricultural system resilience	0.035	
Access to a reliable water supply	0.033	
Governance effectiveness	0.033	

large networks absolute values are expected to be small and, interpreted as a structurally important rank rather than a magnitude. Kumu's Social Network Analysis module (www.kumu.io) was used to analyze the eigenvector values of CLD, as shown in Fig. 5; Table 1.

The higher eigenvector centrality values of barriers indicate structural embeddedness in feedback and are linked to other highly connected barriers within the IWRM operational system network in Ethiopia. The results highlight that household income has the highest eigenvector value (0.062), indicating its central position within the dominant feedback structure linking access to reliable water supply to economic water scarcity. Investing in Water Infrastructure (0.058), equity, and sustainability of water and land development (0.058) reflects their strong integration, with high connective barriers including infrastructure, governance, and effective resource use. Food availability and access (0.057) indicated high centrality, reinforcing its embeddedness at the connection between agricultural yield, water availability, and household income level. The

challenges of placing less emphasis on water development than on management and the inadequacy of hydraulic infrastructure, such as irrigation, storage infrastructure, dams, reservoirs, and water-harvesting infrastructure, manifested in “Economic water scarcity” (0.044).

Transport infrastructure quality, waste infrastructure capacity, equity of water access, and agricultural system resilience (0.035–0.037) indicate their participation in multiple feedback loops. Finally, access to a reliable water supply and governance effectiveness (0.033) had slightly lower eigenvector values; however, they were structurally important because of their consistent connections with highly central nodes.

Thus, a structural analysis based on eigenvector centrality was integrated with a descriptive feedback-loop analysis to link quantitative network metrics with qualitative system interpretation. Eigenvector centrality is used to identify structurally influential variables connected to other highly connected variables within the network. The reinforcing income-infrastructure-water security loop (Fig. 3) was selected for a detailed discussion because it incorporates four of the variables within the highest eigenvector centrality scores, indicating that it represents a structurally significant pathway within the system.

This loop connects household income level, investment in water infrastructure, investment in hydraulic infrastructure, water storage capacity, economic water scarcity, and access to a reliable water supply, thereby capturing the core development water security dynamics. Higher household income levels increase public revenue and the potential for cost recovery, thus enabling greater investment in water infrastructure. Increased investment supports the expansion of the hydraulic infrastructure, including dams, reservoirs, and conveyance systems. These investments enhance the water storage capacity and

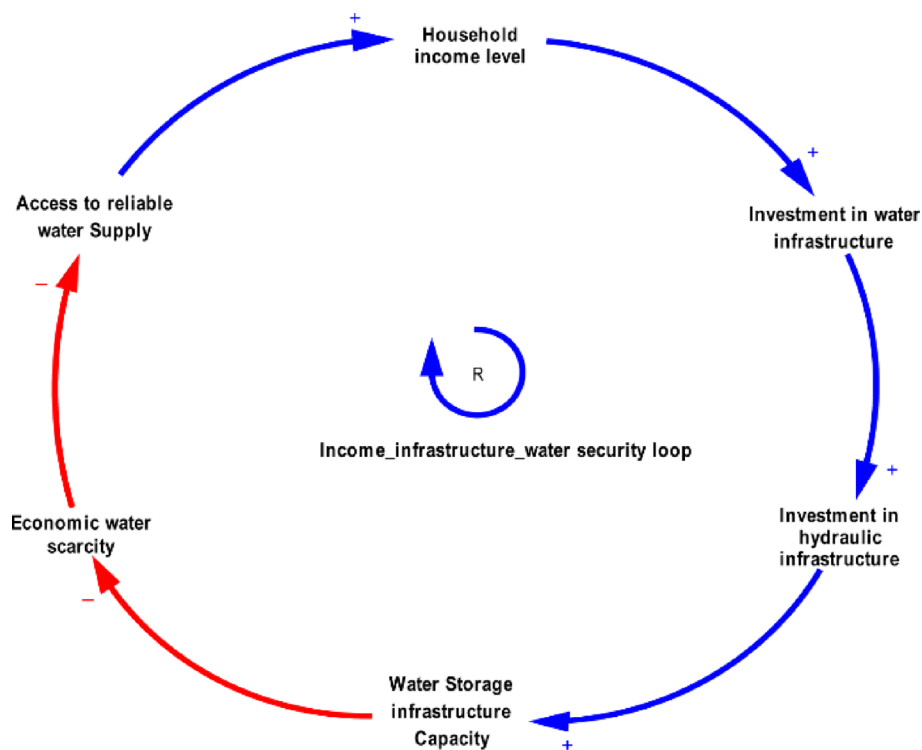


Fig. 3 Household income level, infrastructure, and access to reliable water supply Reinforcing feedback loop in IWRM operation system, CLD

improve the ability of the system scope with seasonal and climatic variability. A high storage capacity reduces economic water scarcity by alleviating supply constraints and lowering shortage-related costs. Reducing economic water scarcity improves access to reliable water supply [10] and, strengthens household livelihoods and productivity. This increases household income levels, thereby reinforcing the feedback structure.

Thus, each barrier of the loop amplifies the other and emphasizes the necessity of targeted interventions to disrupt the cycle and promote sustainable operationalization of IWRM.

3.3 MICMC influence-dependence score and the leverage barriers

MICMAC influence-dependence analysis classifies variables based on their relative positions within the influence-dependence space defined by the selected variables and assumed causal relationships. Therefore, the resulting classification of leverage points is contingent on the system boundaries, variable sets, and causal assumptions used in this study. Thus, the MICMAC method was automatically applied to all key barriers driving system outcomes using Kumu's Social Network Analysis module (www.kumu.io). Dependence and influence scores were calculated (Supplemental Materials Information) and projected onto a two-dimensional chart to classify barriers as shallow leverage points (target), deep leverage points (driver), monitor, and incidental [33, 34, 44, 59] barriers.

Based on the relative influence-dependence positioning, six barriers were classified as deep leverage points within this system configuration, characterized by high influence and high dependency within the IWRM system. This variable structurally occupies central positions in the causal network, participates in multiple feedback loops, exerts strong control over system behavior, and is sensitive to changes elsewhere in the system.

The barriers located in the right quadrant of the chart (Fig. 5) are deep leverage points that, if disturbed, may propagate sequential and interconnected effects across the IWRM operational system. The dominance of deep leverage points may depend on the barriers identified in the analysis, as well as strong interlinking and mutual dependence.

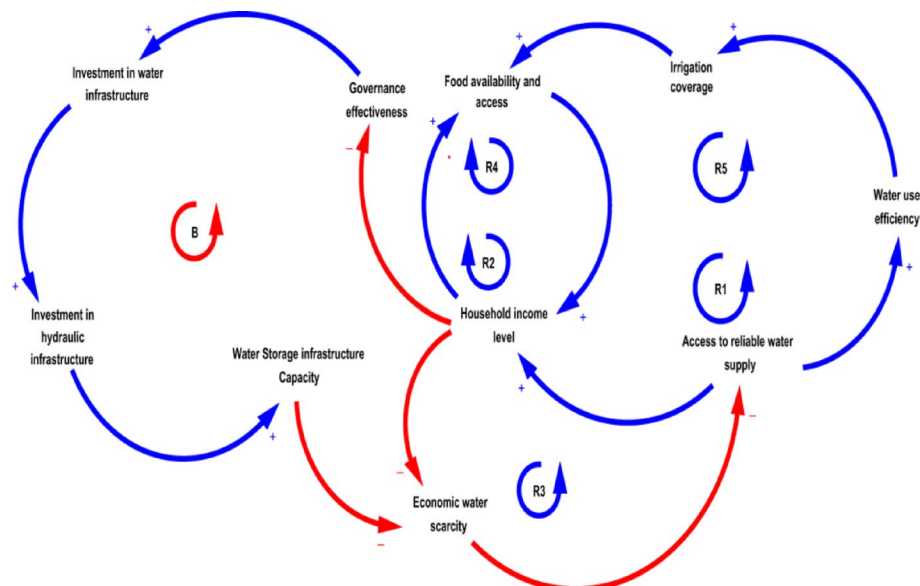


Fig. 5 Causal feedback loops for the highest dominant loops for each leverage point

Household income level
Access to reliable water supply
Governance effectiveness
Economic water scarcity
Food availability and access
Irrigation coverage

Budget allocation for the water sector
Planning and management effectiveness
Infrastructure maintenance capacity
Economic resilience (GDP performance)
Hydrological data availability and quality
Dam and reservoir storage capacity
Adaptive capacity to climate variability

Agricultural system resilience
Equity and sustainability of water and land development
Equity of water access
Investment in hydraulic infrastructure
Investment in water infrastructure
Prioritization for water development
Transport infrastructure quality
Waste management infrastructure
Water pollution
Water quality and quantity degradation

The remaining 54 barriers are incidental

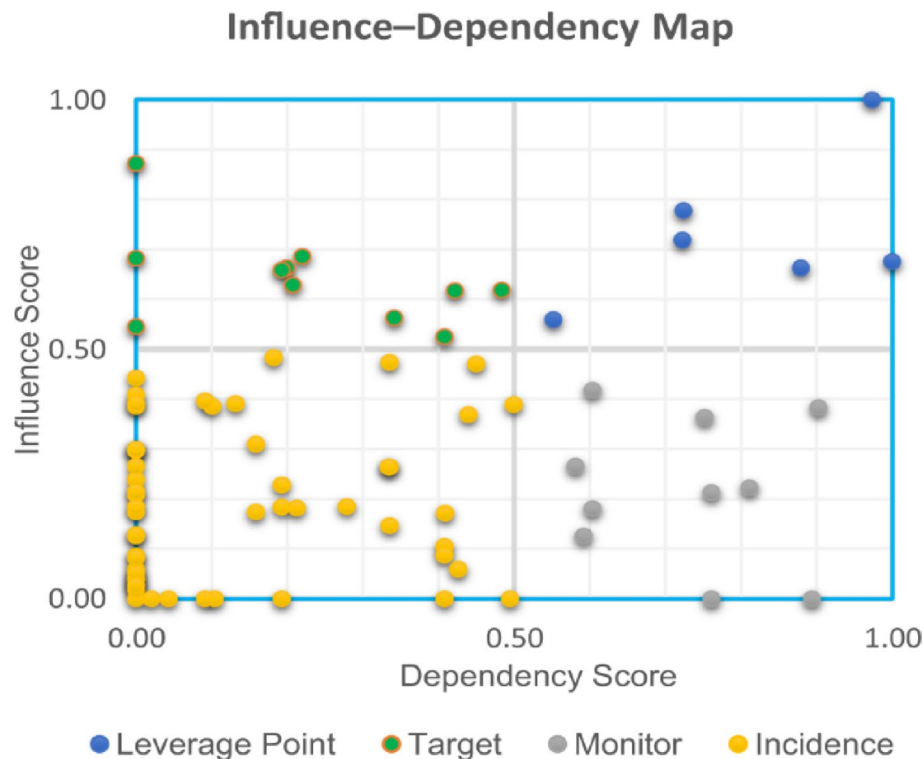
Barriers roles in IWRM System operations

3.4 Leverage points and their feedback loops

The normalized loop-gain values rank the feedback loops relative to one another within the modeled system. A loop with a normalized score of one indicates the relative

Table 2 Key leverage points and MICMAC influence-dependence score

Leverage points	MICMAC-influence	MICMAC-dependence
Household income level	1.000	0.973
Access to reliable water supply	0.778	0.724
Governance effectiveness	0.719	0.722
Economic water scarcity	0.675	1.000
Food availability and access	0.663	0.879
Irrigation coverage	0.559	0.552

**Fig. 4** Role of key variables (barriers) to IWRM in Ethiopia

structural dominance among the identified loops in the model, rather than exclusive control over the system behavior. Accordingly, the structurally dominant reinforcing and balancing loops are interpreted as priority feedback structures within the current causal configuration.

Thus, the analysis of feedback loop participation indicates the differences in how leverage points are embedded within the system structure. A total of 477 feedback loops for the six leverage points were identified from the system structure developed in Vensim PLE, as summarized in Table 2. The household income level is the most structurally engaged leverage point, involved in 126 feedback loops, and is almost evenly classified into balancing (65) and reinforcing (61) loops. This illustrates its central role in amplifying the development path and regulating system stability. Economic water scarcity and governance effectiveness are also highly embedded, with each participating in more than 80 feedback loops that reflect their dual role in driving and constraining water system behavior.

Access to reliable water supply and food availability are involved in more reinforcing than balancing loops, indicating their tendency to reinforce or balance the system once

Table 3 Feedback loops constructed by leverage points

Leverage points	Number of balancing loops (B)	No of reinforcing loops(R)	Total loops
Access to reliable water supply	31	40	71
Household income level	65	61	126
Governance effectiveness	42	39	81
Economic water scarcity	42	42	84
Food availability and access	34	42	76
Irrigation coverage	21	18	39

Table 4 Dominant feedback loops, their weighted score, and normalized score

Loop ID	Loop name	Type	Weighted score	Loop length	Average weighted per variable	Normalized score
R4	Food-income loop	R	0.100	2	0.050	1
R3	Scarcity-access-income loop		0.118	3	0.039	1
R5	Irrigation-food-income-scarcity-access-efficiency loop	R	0.188	6	0.031	1
R2	Income-food security loop	R	0.062	2	0.031	1
R1	Access-efficiency-food-income- scarcity loop	R	0.168	6	0.028	1
B1	Governance-infrastructure-regulation loops	B	0.182	7	0.026	1

activated. In contrast, irrigation coverage participates in fewer loops, suggesting a more targeted but structurally influential role within agricultural feedback structures. Finally, the balance between the reinforcing (242) and balancing (235) loops suggests that Ethiopia's IWRM system simultaneously supports the development process while regulating system stability.

3.5 Dominant feedback loops and systemic influence

The analysis of structurally dominant feedback loops in the IWRM operating system identified six relatively high dominance loops, based on the structural–dynamic dominance analysis described in the methodology section. These loops represent the relative structural dominance of the feedback structures that exert the greatest influence on overall system behavior. Their identification was guided by the combined metrics of eigenvalue centrality, MICMAC influence-dependence scores, and loop structural gain, which capture both the dynamic leverage and structural connectivity of the feedback processes.

The results in Table 3 demonstrate that these six loops (one balancing and five reinforcing) encompass a complex interaction between the six key leverage points. The predominance of the five reinforcing loops in the dominant set highlights several self-amplifying mechanisms that perpetuate the systemic challenges. For instance, loops such as *Income-food availability and access reinforcing Loop* illustrate the mutual reinforcement between income and food availability and access. Increasing household income improves food availability and access by enhancing purchasing power, whereas improved food availability supports labor productivity and economic participation, further increasing income. These findings highlight the strong integration of water and, food systems, with household welfare. Similarly, scarcity-access-income reinforcing loops underscore the feedback between economic water scarcity and income. Increasing economic scarcity limits reliable access to water and reduces income-generating opportunities, thereby

exacerbating economic water scarcity. On the other hand, reducing economic water scarcity improves access and income and reinforces direct outcomes.

By contrast, the identified balancing loop represents a regulatory mechanism of governance, infrastructure, and income capacity. Improved governance effectiveness enables increased investment in water and hydraulic infrastructure, thereby expanding water storage capacity and household income as income levels rise, expectations, service demands, and institutional pressure increase, potentially constraining governance effectiveness if an institution simultaneously increases its capacity.

Therefore, the imbalance between reinforcing and balancing feedbacks explains why efforts to implement IWRM in Ethiopia often face infrastructural, institutional, and financial constraints despite policy interventions.

3.6 Causal loop diagram of dominant feedback loops

Finally, all rankings and classifications are sensitive to the model system structure, including variable selection, assumed causal directions and polarities, and exclusion of time delays and non-linear dynamics. Thus, the result represents the relative behavior of the modeled system and may vary under alternative structural assumptions or expanded variables.

4 Discussions

The integration of eigenvector centrality, MICMAC influence-dependence classification, and dominant feedback loop analysis to develop an interpretation of the system structure beyond direct causal interactions and to move beyond the static representation of Ethiopia's IWRM operation system towards a dynamic, policy-relevant interpretation of system behavior. Eigenvector centrality identifies variables that are structurally embedded, directly or indirectly, with other influential variables within the system, whereas MICMAC identifies the target, leverage point, monitor, and incidence variables based on their relative influence and dependence within the causal network. The integration of these two metrics enables the robust identification of systematically important variables that not only exert a strong influence but are also located within a dominant feedback structure, as identified through loop-gain assessment and recurrent across leverage points, revealing how highly connected variables actively participate in reinforcing and balancing dynamics. The dominance of reinforcing and balancing feedback refers to their relative prevalence and cumulative influence strength within the overall structure compared with other pathways. This integration approach allows the discussion to move beyond variable ranking to understand how strategic leverage points interact dynamically to shape system behavior, stability, and potential interventions within Ethiopia's IWRM operational system.

4.1 Leverage points, dominant feedback loops, and systemic leadership in IWRM

Variables such as household income level, access to reliable water supply, food availability, economic water scarcity, irrigation coverage, and governance effectiveness are structurally embedded within the dominant feedback loops. These variables were also classified as leverage points in the MICMAC analysis, confirming their dual role as highly influential for system connectivity and robust priority intervention points.

4.1.1 Household income level and income–food security livelihood reinforcing loop (R2)

Household income level exerts a strong influence within the CLD structure, with the highest eigenvector value (0.062), maximum MICMAC influence (1.000), and high dependency (0.973), indicating a systemic leverage point. Its central position reflects bidirectional linkages among water access, food availability, infrastructure investment, and governance outcomes. The causal interactions in CLD Fig. 1 indicate that higher household income enhances food availability and access, improves the quality of transport infrastructure, and enables greater investment in water infrastructure, while simultaneously reducing economic water scarcity. This aligns with Ethiopia's water and economic development context, where livelihoods are highly vulnerable to rainfall variability, limited irrigation access, and water scarcity, and increasing income reduces vulnerability to economic water scarcity [7, 19]. More than 80% of Ethiopia's population relies on rainfed agriculture, making rural livelihoods extremely vulnerable to climate variability and recurrent drought due to insufficient irrigation and low agricultural productivity [61–64]. Low agricultural productivity reduces GDP and household income, limits the capacity to invest in and access water infrastructure, and hinders the implementation of adaptive measures, resulting in socioeconomic disparities [65, 66]. For instance, according to Borgomeo et al. [68] and Shah [13], a decrease in rainfall could lead to a 5% decrease in the GDP of the Awash Basin, which is more severe than a 10% decline in agricultural GDP [13, 67].

At the same time, household income is reinforced by access to reliable water supply, food availability, and governance effectiveness, generating reinforcing and stabilizing feedback. The high number of incoming and outgoing causal links indicates a higher eigenvector value and confirms that household income is the core transmission variable through which changes in reliable water access, infrastructural investment, and institutional performance are distributed across the IWRM operational system. Despite being one of the fastest-growing economies in the region, Ethiopia remains one of the poorest, with low per capita income and a limited focus on investment in the water sector [68].

The *household income–food access and availability reinforcing loop (R2)* is the shortest, and structurally dominant loop (average weighted score of 0.0312, normalized to 1) and powerful livelihood-driven feedback loop linking household income level and food availability and access through mutually reinforcing interactions (Fig. 3). A higher household income improves food availability and access by enabling greater purchasing power, investment in agricultural inputs, and improved market access. Enhanced food availability and access strengthen household income by improving labor productivity, nutritional outcomes, and the capacity to engage in income-generating activities.

The loop is self-reinforcing, amplifying the initial change in income or food availability, and its high average weighted contribution (Table 3) indicates a strong feedback intensity per variable, thereby magnifying gains or losses. When income and food access improve simultaneously, the loop accelerates livelihood enhancement; In contrast, income shocks or food shortages can trigger a downward spiral of poverty and food insecurity.

Within the IWRM system, R2 functions as a core reinforcing mechanism that couples water-dependent food production with socioeconomic well-being. Its dominance highlights why interventions that improve either food access or household income can generate cascading benefits, while neglecting these dimensions risks reinforcing vulnerability across the system.

4.1.2 Access to reliable water supply and water–food–income reinforcing loop (R1)

Access to a reliable water supply emerges as the second structurally influential leverage point in the Ethiopian water resources management system, with an influence score of 0.778, dependency of 0.702, and relatively medium eigenvector centrality of 0.033, underscoring its central position in multiple feedback loops linking water availability, household income, agricultural resilience, and income generation. However, not the highest structurally central variable, its leverage behavior indicates that improvements in water access activate multiple reinforcing feedback loops.

The CLD in Fig. 1 shows that a reliable water supply directly affects household income levels, equity in water access, water-use efficiency, adaptive capacity to climate variability, and the equity and sustainability of water and land development, while reducing reliance on rainfed agriculture. At the same time, economic water scarcity inversely constrains access to reliable water supply, indicating its sensitivity to upstream economic water scarcity and infrastructural conditions. This, by directional interaction, is a reliable water supply as a critical mediator linking infrastructure investment, livelihood, and resilience.

In Ethiopia, spatiotemporal rainfall variability [10, 21], limited storage infrastructure capacity [10, 13, 67], and climate-induced droughts constrain reliable supply, agricultural production, domestic supply, and economic development [64, 67]. Thus, access to a reliable water supply is not a symptom of climatic variability but a structural barrier that reinforces income and food availability, thereby hindering economic development.

The dominant Access-Efficiency-Food-Income-Scarcity Reinforcing Loop (R1), among the 71 reinforcing loops (Table 2), is the highly dominant feedback loop (average weighted score of 0.028, normalized to 1) and represents a core livelihood-water-agriculture feedback loop that links infrastructure performance, agricultural productivity, and socioeconomic outcomes through a closed causal chain. The loop begins with access to a reliable water supply, which increases water-use efficiency, enabling more effective allocation and better timing of water resources. Improved efficiency expands irrigation coverage, thereby enhancing food availability and accessibility. Improved food access strengthens household income, and reduces economic water scarcity by improving affordability and investment capacity. Among the 71 reinforcing loops (Table 2), the Access-Food-Income-Scarcity Reinforcing Loop, a structurally dominant loop (weighted score of 0.168, normalized to 1), illustrates how economic water scarcity and access to a reliable water supply led to a self-reinforcing cycle.

The reduced economic water scarcity further improved access to reliable water supply illustrating a self-reinforcing cycle. This reinforcing loop is evident across all rural areas of Ethiopia, where low irrigation coverage and seasonal shortages exacerbate poverty and economic vulnerability [19, 24]. Addressing water insecurity for sustainable development requires significant investment in both infrastructure and institutions, particularly in countries with variable hydrology [7]. Thus, breaking this loop requires investing in infrastructure, including small to medium-scale storage, water-harvesting technologies, a diversified water-supply system, and improved watershed management to stabilize water availability and enhance economic resilience [10, 24].

4.1.3 Governance effectiveness and governance–infrastructure–scarcity regulation loop (B)

Governance effectiveness operates as a balancing leverage point in IWRM operational systems, with an influence score of 0.719, a dependence score of 0.722, and a relatively moderate eigenvector centrality of 0.033, which indicates that a structurally embedded yet strongly independent role is a foundational barrier to effective IWRM operations. The CLD shows that improved governance effectiveness strengthens investment in water infrastructure, adoption of modern technologies, enforcement of legal and regulatory frameworks, and marketing institutional capacity, while reducing the policy implementation gap. Conversely, higher household income levels reduce governance pressure, generating stabilizing feedback that moderates the need for institutional interventions. In Ethiopia, fragmented institutional arrangements, poor stakeholder coordination and participation, overlapping responsibilities, limited budget allocation, and weak enforcement capacity constrain effective water resource management [11, 12]. This undermines integrated planning and management, resulting in ineffective IWRM operationalization [11, 69], thereby regulating system behavior through balancing dynamics rather than driving expansion. This indicates that governance critically drives systemic outcomes through hierarchical rather than network-wide connections, indirectly influencing other variables through institutions, and is a structurally central barrier, influenced by multiple barriers in the system. This dual enabling and regulatory role of governance effectiveness shapes system performance without dominating network connectivity, thereby stabilizing the interactions between infrastructure development, equity allocation, and service delivery.

The Governance-Infrastructure Loop (B) functions as structurally balancing loops, indicating regulatory potential of institutional capacity in moderating infrastructure expansion and ensuring reliable water access within the IWRM system. The loop is initiated by governance effectiveness, which enhances policy coordination, planning quality, and budget allocation, thus enabling sustainable investment in the water and hydraulic infrastructure. These investments expand the water storage capacity, improve the ability of the system to overcome hydrological variability, regulate seasonal flows, and distribute reliable water across sectors. As a result, economic water scarcity declines, as financial, institutional, and infrastructural barriers to water access are reduced [10, 21].

Reduced economic water scarcity increases access to reliable water supplies, thereby enhancing household income levels by improving agricultural productivity, reducing climate-induced shocks, and stabilizing livelihoods [7, 19]. However, rising household income exerts an inverse influence on governance effectiveness by reducing crisis-driven institutional pressure and shifting political and policy attention towards other development priorities. This feedback further constrains the acceleration of infrastructure investment, thus closing the loop and stabilizing the system dynamics [9, 12, 16]. Within the IWRM operation system, this balancing mechanism loop underscores the importance of sustaining governance capacity and institutional strength as access to a reliable water supply and socioeconomic conditions improve. Otherwise, improvements in infrastructure and income may adversely affect institutional capacity, thereby limiting the long-term system resilience and adaptive capacity.

4.1.4 Economic water scarcity and scarcity–access–income reinforcing loop (R3)

Economic water scarcity emerges as a dominant leverage point based on relative influence-dependence positioning, characterized by the highest dependence (1.000), moderate influence (0.675), eigenvector centrality (0.045), and participation in multiple reinforcing loops, indicating its sensitivity to upstream system dynamics rather than an independent causal driver. This configuration indicates that economic water scarcity functions primarily as an integrative system outcome, reflecting the cumulative effect of upstream structural constraints rather than the absolute water availability [10].

The Causal link analysis in Fig. 1 shows that economic water scarcity directly constrains access to a reliable water supply and agricultural system resilience, which is exacerbated by reservoir degradation, reservoir water loss, and physical water scarcity. However, higher household incomes and expanded water storage infrastructure capacity have reduced economic water scarcity, improved access, mitigated seasonal variability, and enabled the effective use of available water resources. In contrast, in Ethiopia, limited per capita storage, uneven spatial distribution of water, and inadequate infrastructure investment amplify the effect of rainfall variability on persistent water stress [10, 21].

This dynamic position of economic water scarcity as a critical leverage point for reducing reinforcing scarcity-access-income feedback within the IWRM operation system.

The *scarcity-access-income loop*, R3 (average weighted score of 0.039, normalized to 1), describes a self-amplifying loop between economic water scarcity, access to a reliable water supply, and household income, which reduces access to reliable water by limiting households and institutional capacity to invest, operate, and maintain water infrastructure and services. Reduced access to water limits productive water use for agriculture and livelihoods, leading to lower household income and development [7, 10]. Lowering income further intensifies economic water scarcity by limiting households' capacity to invest in water storage, irrigation, and service payments, thereby reinforcing scarcity conditions [21].

Although the loop contains two inverse causal links, their combined effect produces overall reinforcement, amplifying either scarcity or recovery depending on the initial condition. When scarcity increases, the loop drives a downward spiral of reduced access and declining income. Conversely, improvements in income or water access can reduce scarcity pressure. Within IWRM implementation, R3 highlights the structural link between water access and economic capacity, illustrating that infrastructure-led interventions alone are insufficient unless accompanied by livelihood and investment capacities. Therefore, addressing this reinforcing loop is critical for breaking income-scarcity traps and improving both water security and socioeconomic resilience.

4.1.5 Food availability and access and food–income mutual reinforcement loop (R4)

Food availability and access are structurally important leverage points within the modeled system, with relatively high eigenvector centrality (0.057), an influence score of 0.683, and a strong dependency of 0.879, reflecting its deep embedding within livelihood-water-agriculture feedback.

The CLD analysis indicates that improved feed availability increases household income, a finding reinforced by irrigation coverage, agricultural yields, farmer income, and market accessibility. This bidirectional linkage generates reinforcing feedback loops

that link food security and income dynamics tightly. In Ethiopia, the dominance of rain-fed agriculture, recurrent drought, and limited irrigation expansion amplify the sensitivity of food availability to climatic variability, particularly in lowland and drought prone area [24, 30].

The Food-Income Loop (R4) reflects bidirectional livelihood dynamics in which food availability and household income mutually reinforce one another. Improvements in food availability enhance household income by increasing labor productivity, nutritional status, and agricultural market participation. In turn, a higher household income improves food availability and access by increasing purchasing power, investing in farming inputs, and expanding market access. This loop is structurally strong because it links the two high-centrality livelihood variables identified in the eigenvector and MICMAC analyses.

Thus, for IWRM implementation, R4 acts as a core reinforcing livelihood loop, strongly linking water-dependent food systems with socioeconomic outcomes, and underscores that water interventions influence food production, cascading socioeconomic impacts across interconnected sectors.

4.1.6 Irrigation coverage and irrigation-food-income-water efficiency reinforcing loop (R5)

Irrigation coverage shows the lowest eigenvector centrality among the leverage points (0.021); however, it has medium influence (0.559) and dependency (0.552), confirming its leverage role despite its limited structural connectivity. The causal interaction shows that expanding irrigation coverage improves food availability and access and enhances the utilization of water and land resources, while being enabled by irrigation infrastructure, water-use efficiency, and water storage capacity. In Ethiopia, the underutilization of irrigation potential despite extensive suitable land for agricultural production is highly exposed to rainfall variability and recurrent droughts [19, 24]. Consequently, irrigation coverage reinforces feedback loops linking climate variability, food insecurity, and household income, indicating that its leverage effect arises from its position within agricultural-food-income dynamics rather than from extensive network centrality.

The Irrigation–food–income–water efficiency reinforcing loop (R5) integrates irrigation expansion with Water Efficiency and scarcity dynamics. Expanded irrigation coverage increases availability, which, in turn, increases household income. Higher income reduces economic water scarcity by enabling investments in storage, services, and technology. Reducing economic water scarcity improves access to a reliable water supply, enhances water-use efficiency, and supports further irrigation expansion. This loop links technical performance with socioeconomic capacity, thereby, creating a cumulative pathway for agricultural intensification.

From an IWRM perspective, R5 demonstrates that irrigation development is most effective when embedded within efficiency and affordability frameworks. Without such integration, expansion risks increase demand pressures; thus, irrigation becomes a driver of sustainable water-food-income interactions [7, 13].

4.2 Implications of leverage points and dominant loops for IWRM operationalization

The analysis of six leverage points reveals that constraints on effective IWRM operationalization in Ethiopia arise from structurally embedded reinforcing and balancing feedback rather than from isolated sectoral limitations. Integrated MICMAC, eigenvector

centrality, and feedback loop analyses indicate that IWRM performance depends on the activation leverage points located at the intersection of infrastructure, livelihoods, and institutions. Structurally central variables, such as household income level, food availability, and access, function as system connectors, transmitting the effects of water investments across agricultural productivity, economic resilience, and social equity. In contrast, unreliable water supply and economic water scarcity translate hydrologic variability into reinforcing cycles of low productivity, livelihood vulnerability, and constrained adaptive capacity, particularly under conditions of limited storage and uneven spatial distribution of water resources [10, 13, 21]. Their participation in multiple reinforcing loops (R1, R2, R4, and R5) indicates that improvements in water reliability, irrigation coverage, and efficiency can yield cumulative livelihood gains when affordability and access constraints are concurrently addressed.

Although Ethiopia has substantial annual surface runoff [19, 63, 70], its water storage capacity remains limited. Current surface storage accounts only approximately 8% (3.28 km³) of the estimated Seasonal Storage Index of 40.99 km³, and reservoir storage per capita remains below 1% of the potential capacity [10, 21]. At the same time, the irrigation potential remains underutilized, with only approximately 20% of suitable land being developed, leaving agricultural productivity highly exposed to rainfall variability and recurrent drought [64, 71]. Limited water infrastructure and inefficient management systems further exacerbate economic water scarcity, particularly during seasonal and climatic fluctuations [10, 18, 72]. Budget allocations to the water sector have remained relatively low and inconsistent, fluctuating between 1.9% and 4.5% of the federal capital budget, despite increasing population pressure and sectoral water demand, constraining infrastructure expansion and institutional capacity development [10, 12]. Institutional constraints, including overlapping mandates, weak enforcement, fragmented basin coordination mechanisms, and limited hydrological data systems, further weaken the balancing governance infrastructure loops [10, 12]. Additionally, rapid urbanization, industrial discharge, and agricultural runoff have degraded the water quality in major basins, reducing effective water availability and increasing economic water scarcity [9, 26]. In general, these hydro-climatic, infrastructural, financial, and institutional dynamics reinforce scarcity-access-income feedback and weaken stabilizing governance loops, aligning empirical trends with the dominant feedback structure identified in the model. This highlights that addressing water scarcity in Ethiopia requires not only infrastructural expansion, but also strengthened coordination, improved water management, and integrated policy interventions that consider both hydrological and socioeconomic realities.

Building on these systemic constraints, leverage points with moderate structural centrality but strong relative influence-dependence characteristics, particularly access to reliable water supply, governance effectiveness, economic water scarcity, and irrigation coverage, play a crucial role in shaping system behavior. These points can regulate or amplify the intersection of infrastructure, institutional capacity, and socioeconomic factors. Balancing loops centered on governance and infrastructure investment illustrate the potential of institutional capacity to stabilize system expansion, preventing short-term gains from being involved in long-term inefficiencies.

However, weak coordination, limited budget allocation, and implementation gaps constrain this regulatory function, allowing reinforcing scarcity loops to dominate [9, 12]. This aligns with the evidence that implementing IWRM in water-insecure countries

with high rainfall variability and complex hydrology, without adequate storage, infrastructure, and institutional resilience, faces persistent challenges [5, 13, 21, 63, 7]. Similarly, successful operationalization of IWRM in Ethiopia requires the adaptation of interventions to local contexts [5].

Breaking these reinforcing scarcity loops requires integrated interventions that simultaneously address infrastructure development, governance capacity, and livelihood vulnerability. Empirical evidence shows that expanding multipurpose storage, groundwater recharge, and decentralized rainwater harvesting enhance hydrological reliability and climate resilience [10, 13, 19, 21, 23, 31, 73–75]. Modernizing irrigation systems, rehabilitating canals, and adopting efficient technologies such as drip and sprinkler irrigation can improve water productivity and food security [23, 31]. Strengthening basin authorities, improving hydrological and administrative data systems, and leveraging public-private partnerships can reduce coordination failures and mobilize sustainable investment. At the community scale, solar-powered irrigation, micro-power systems, and small-scale storage ponds directly enhance income levels and food availability by improving access to rural water. Constructed wetlands and water reuse systems further contribute to pollution control and maintenance of environmental flow [74, 76–78].

Overall, the dominance of reinforcing and balancing feedback demonstrates that IWRM performance improves when policy interventions align with infrastructural development, institutional capacity, and socioeconomic resilience, enabling dynamics while constraining scarcity-driven traps at both the national and basin scales.

The identified feedback structure confirms that IWRM challenges in Ethiopia are inherently context-specific and, rooted in national arrangements, resource allocation mechanisms, and socioeconomic conditions. Consequently, the universally applied IWRM solutions are unlikely to be effective. Instead, the identified leverage points, particularly governance effectiveness, coordinated infrastructural investment, and livelihood water linkages, align with Ethiopia's national water policy objectives while revealing persistent implementation constraints. Addressing these constraints required adaptive and coordinated policy responses that explicitly engage with the reinforcing and balancing dynamics embedded in the IWRM system.

In this way, the study contributes to SDG 6.5.1 by demonstrating that the degree of IWRM implementation depends on governance effectiveness, institutional coordination, and investment capacity, which jointly condition equitable and reliable water access beyond service coverage indicators alone.

5 Conclusion

This study demonstrates that the operationalization of Integrated Water Resources Management (IWRM) in Ethiopia is constrained not by isolated technical limitations but by structurally embedded interactions among hydrological variability, limited infrastructure, institutional capacity, and socioeconomic vulnerability. Unlike traditional sectoral water management approaches, IWRM seeks to coordinate water, land, and related resources to achieve economic efficiency, social equity, and environmental sustainability. The findings indicate that this coordination challenge is inherently systemic, and, shaped by a feedback structure that cannot be addressed through generic or blueprint solutions.

By integrating MICMAC influence-dependence analysis, eigenvector centrality, and feedback loop analysis within a systems thinking framework, this study identifies six

leverage points and the associated dominant reinforcing and balancing feedback loops that shape IWRM implementation performance. Household income level, food availability, and access function as system-wide connectors, while access to reliable water supply, governance effectiveness, irrigation coverage, and economic water scarcity operate as strategic intervention nodes. These results demonstrate that leverage points are defined not by structural centrality alone, but by their participation in dominant feedback mechanisms that translate hydrological variability into socioeconomic outcomes.

The structure of the identified feedback loops further underscores the context-specific nature of the IWRM outcomes in Ethiopia. Reinforcing loops linking water access, income, food availability, and water scarcity can generate persistent income-food-scarcity traps, particularly under conditions of rainfall variability and limited storage capacity. By contrast, balancing loops associated with governance and infrastructure investment provide potential regulatory pathways, but their stabilizing effect is weakened by institutional fragmentation and financial constraints. This highlights the need for context-adaptive implementation strategies that respond to national administrative arrangements, economic conditions, and hydrological realities.

These findings are directly relevant to Ethiopia's national water policy objectives, which emphasize equitable access, integrated planning, and resilience to climate variability. While these objectives align with the IWRM principles, the persistence of reinforcing scarcity loops indicates that implementation, rather than policy intent remains the principal challenge. Therefore, strengthening institutional coordination, prioritizing sustained investment in storage and irrigation infrastructure, and aligning governance reforms with livelihood improvement strategies are, critical for translating policy commitments into measurable outcomes.

In general, the study indicates that improving IWRM performance requires shifting from sector-specific interventions to coordinated strategies that explicitly target leverage points and feedback structures.

Future research could extend this work by integrating dynamic system modeling and spatial analysis using tools such as the Water Evaluation And Planning (WEAP) model, combined with stakeholder based scenario testing. This approach would enable the evaluation of long-term interventions, refinement of context-specific strategies, and translation of structural insights into actionable, practical solutions to enhance IWRM performance.

By doing so, Ethiopia and other countries facing similar hydro-climatic and institutional conditions can progress towards more resilient, equitable, and adaptive water management systems consistent with the objective of Sustainable Development Goal 6.5.1.

Supplementary Information

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Supplementary Material 1

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Author contributions

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Data availability

The data supporting the findings of this study are included within the article and its supplementary information.

Declarations**Ethics approval and consent to participate**

The study involved consultations with professional experts regarding the integrated water resources management in Ethiopia. The research did not involve experiments on human or animals and did not include the collection of personal, sensitive, or biological data. Therefore, formal ethics committee approval was not required. Prior to each discussion, the purpose of the study was clearly explained to the participants, and their voluntary agreement to participate was obtained before continuing the discussion.

Consent for publication

Not applicable. The study did not include identifiable personal data from participants or experts.

Competing interests

The authors declare no competing interests.

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